Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

1. Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ Retrieve the values of $P_{u,t}$

$P_{u,t} \leftarrow \texttt{Power.GetValue}(t)$

end for

for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ Initial conditions on the state variables

if $P_{u,t} > \texttt{MinimumPower.GetValue}(t)$ **then**

$S_{u,t}^{\text{OFF}} \leftarrow 0$

$S_{u,t}^{\text{STOP}} \leftarrow 0$

$S_{u,t}^{\text{START}} \leftarrow 0$

if $P_{u,t} < P_{u,t-1}$ **then** ▷ Exact initialization required only if the FLAT state is defined.

$S_{t-1}^{\text{ON_UP}} \leftarrow 0$

$S_{t-1}^{\text{ON_DOWN}} \leftarrow 1$

$S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$

else if $P_{u,t} > P_{u,t-1}$ **then**

$S_{t-1}^{\text{ON_UP}} \leftarrow 1$

$S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$

$S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$

else if $P_{u,t} = P_{u,t-1}$ **then**

$S_{t-1}^{\text{ON_UP}} \leftarrow 0$

$S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$

Algorithm 1: Initialization of the control and state variables